

# Fuel Subsidy Removal in Nigeria: Household Welfare and Public Opinion \*

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## Abstract

**Motivation:** In May 2023 Nigeria removed its fuel subsidy, raising pump prices overnight in a country where most people are poor and most electricity is generated privately from petrol and diesel. The consequences for households, and how households judged the reform, have not been documented at national scale.

**Purpose:** This paper assesses how Nigerian households fared after the removal across several dimensions of welfare, how aware of and supportive of the reform they were, and what both imply for the design of future subsidy reform.

**Approach and Methods:** I use repeated observations from the Nigeria National Longitudinal Phone Survey spanning the reform. I report descriptive before-and-after changes in six self-assessed welfare measures, a difference-in-differences design that exploits pre-determined geographic variation in fuel distribution infrastructure, and household responses to a subsidy opinion module.

**Findings:** Welfare fell after the reform on five of six dimensions, food, clothing, healthcare adequacy, living standards, and happiness, with only housing unchanged. The decline was still present fourteen months later. States more integrated into formal fuel markets, which are also more economically developed, recorded smaller declines. Of households that expressed a view, 87 percent disapproved, most citing higher prices, while the minority who approved gave the standard economic arguments against subsidies.

**Policy Implications:** Compensation should reach households before pump prices rise and should be sustained. Nigeria's transfer programme had reached about a third of its target households two years after the reform. Transfers should be weighted toward the least developed states, which absorbed the largest welfare declines. The government should publish what the reform saved and where the money went, since the public accepted the price increase without seeing the fiscal return, and should lower the fuel intensity of household consumption through electricity and transport investment.

**Keywords:** fuel subsidies, Nigeria, policy reform, household welfare, public opinion.

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# 1 Introduction

On 29 May 2023, in his inaugural address, Nigerian President Bola Tinubu announced that the Federal Government would immediately stop subsidizing petroleum products. The price of petrol, kerosene, and diesel rose the following day (Omotayo, 2024).<sup>1</sup> The fiscal rationale for the decision was straightforward. The International Monetary Fund estimated that implicit fuel and electricity subsidies would consume 3 percent of GDP if maintained (Olawin, 2024). No compensatory programme accompanied the announcement, so households absorbed the price increase without offsetting support (Chukura, 2024).<sup>2</sup>

The reform landed on a population with little margin to absorb it. More than half of Nigerians are classified as poor, and roughly 30 percent live below the international poverty line.<sup>3</sup> Electricity provision is erratic (Oladosu & Adegbulugbe, 1994), and about three-quarters of electricity consumption is generated privately through petrol or diesel generators (Nnodim, 2023). A change in the fuel price therefore changes more than the cost of transportation. It changes the cost of keeping the lights on, running a shop, pumping water for a farm, and bringing food to market.

This paper asks how the removal affected households, how households themselves understood it, and what the two together imply for how such a reform should be designed. The paper makes four contributions. First, it documents the welfare consequences across six self-assessed dimensions, food, clothing, housing, healthcare adequacy, living standards, and happiness, on a nationally representative panel observed before and after the reform. Welfare fell on five of the six dimensions. Existing studies of the 2023 removal document adverse effects within single regions, the Southeast and the Northwest (Datti, 2024; Meludu, Komolafe, & Chilaka, 2024); this paper measures household welfare nationally and across these six dimensions at once. Second, it brings household perceptions to the centre of the analysis. About half of house-

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<sup>1</sup>The pump price of petrol rose from roughly 185 naira per litre before the reform to about 600 naira within two months, a record at the time (Al Jazeera, 2023).

<sup>2</sup>Advocates of subsidy removal stress the role of government transfers in cushioning the immediate effects (Siddig, Aguiar, Grethe, Minor, & Walmsley, 2014).

<sup>3</sup>World Bank, *Nigeria Poverty and Equity Brief* (World Bank, Washington, DC, 2023).

holds were aware of the subsidy, and of those who offered a view, the overwhelming majority disapproved of its removal. The reasons households gave reveal a gap between the public's objection, which rests on the price increases they experienced, and the efficiency-and-fairness case that a small minority of supporters echoed. Third, it provides suggestive evidence on how the burden varied with integration into formal fuel markets, using a difference-in-differences design built on geographic variation in fuel distribution infrastructure. Fourth, it converts these results into a set of design recommendations for subsidy reform, covering the sequencing of compensation, the geographic weighting of transfers, the disclosure of fiscal savings, the fuel intensity of household consumption, and the pricing rule. Each recommendation is tied to a specific finding, and I set out in each case what the evidence here can and cannot support.

This paper does not claim to identify the average causal effect of the reform. Every household experienced the same nationwide price shock alongside a concurrent currency devaluation, so no untreated comparison group exists. The before-and-after estimates are descriptive, and the difference-in-differences estimates recover differences across more- and less-exposed areas rather than the level of the effect. The infrastructure measure I use is correlated with economic development, and I treat the heterogeneity it reveals in that light. This paper presents a clear account of what happened to households, what they thought of it, and how the burden was distributed, on representative data spanning the reform.

The paper proceeds as follows. Section 2 sets out the political economy of fuel subsidies, beginning with why governments subsidize and why subsidies generate welfare losses, then turning to the evidence on fuel subsidies and to the Nigerian case in depth. Section 3 describes the data and research design. Section 4 presents the results. Section 5 discusses the results, sets out their policy implications, and concludes.

## 2 Political Economy of Fuel Subsidies

### 2.1 Why Governments Subsidize

Governments subsidize goods for three broad reasons. The first is to correct a market failure. When the private benefit of consuming a good falls short of its social benefit, private consumption settles below the socially desirable level, and a subsidy can close the gap (Pigou, 1920). The second reason is distributional. In countries without a developed system of cash transfers, holding down the price of a widely consumed good is an administratively simple way to raise the real incomes of the poor. The third is political. A subsidy creates a visible, broadly shared benefit, and its removal imposes a visible, broadly shared cost, which makes subsidies attractive to incumbents and difficult to withdraw once granted.

Fuel subsidies in resource-rich states carry a further rationale. Citizens of an oil exporter expect to share in the resource wealth, and cheap petrol is a direct and legible form of that sharing. Where the state delivers few other services, the fuel subsidy becomes one of the few tangible returns the public receives from oil (Ozili & Obiora, 2023). This logic helps explain why fuel subsidies, often introduced as temporary measures, tend to persist, and why attempts to remove them provoke sustained opposition.

### 2.2 Why Subsidies Generate Welfare Losses

A subsidy that lowers the price a consumer pays below the cost of supplying the good drives a wedge between the two. Consumers buy more than they would at the undistorted price, up to the point where the value of the last unit consumed falls below the cost of providing it. The difference is a deadweight loss: resources supply fuel worth less to its users than it cost to import or refine (Auerbach & Hines Jr, 2002). The loss grows with the *square* of the distortion, so a subsidy of the size Nigeria maintained produces a loss out of proportion to it (Harberger, 1964). Conventional estimates understate the cost because they omit behavioural responses such as smuggling and

black-market arbitrage (Feldstein, 1999), both features of Nigeria's subsidized market.

A subsidy is also a poor instrument for the distributional purpose it is meant to serve. The same money delivered as a targeted cash transfer raises the welfare of the poor by more, because it relieves the household budget without distorting what the household buys (Atkinson & Stiglitz, 1976). Demand for fuel changes little when its price falls, so most of the outlay substitutes for spending households would have undertaken anyway, while the efficiency cost remains (Parry & Small, 2005; Ramsey, 1927).

The rents a subsidy creates also attract resources devoted to capturing them (Krueger, 1974; Tullock, 1967). Nigeria's subsidy regime generated fraudulent claims for fuel never delivered, cross-border smuggling to higher-price neighbours, and patronage networks organized around the distribution of subsidized product (Gboyega, Soreide, Le, & Shukla, 2011).

## **2.3 Fuel Subsidies in Practice**

The case against fuel subsidies rests on evidence as well as theory. Fuel subsidies are regressive. Across developing countries, the richest fifth of households captures around 43 percent of the benefit while the poorest fifth receives about 7 percent (Arze del Granado, Coady, & Gillingham, 2012). The Nigerian case matches this: the top fifth of households enjoys roughly twice the benefit of the bottom fifth (Soile & Mu, 2015). This is because wealthier households own more vehicles, run more generators, and consume more fuel-intensive goods, so a price held artificially low transfers more to them in absolute terms. Fuel subsidies also encourage consumption of a good whose social cost exceeds its private cost; removing them reduces energy use and emissions and shifts resources toward more productive uses, even in oil exporters (Glomm & Jung, 2015; Plante, 2014).

The difficulty with fuel subsidy removal in Nigeria lies in the transition. The international experience suggests that reforms succeed when they are clearly communicated, phased in, paired with an automatic pricing mechanism, and accompanied by

support for the poor (Clements et al., 2013). Nigeria's May 2023 reform met few of these conditions. It was abrupt, announced in a single sentence, and unaccompanied by social protection. Recent Nigerian studies record the consequences: removal raised food prices and worsened food insecurity in the Southeast (Meludu et al., 2024) and, among urban households in the Northwest, shifted spending toward transport, food, healthcare, and energy while reducing savings (Datti, 2024).

## 2.4 The Nigerian Case

Nigeria combines the general features of fuel subsidies with conditions that magnify the effect of removal. It is Africa's most populous country and one of its largest oil producers, yet more than half its people are poor and about 30 percent live below the international poverty line (World Bank, 2022). Petroleum accounts for roughly 58 percent of GDP (Gboyega et al., 2011). Of the 36 states in the country, only nine in the south produce oil, so the revenue generation is concentrated geographically while the subsidy that returned some of it to the public was spread nationwide.

A structural feature of the Nigerian oil sector shapes our understanding of this issue. Despite exporting crude, Nigeria refines little of it at home. State-owned refineries have operated far below capacity for decades, so the country exports crude and imports finished petroleum products for domestic use.<sup>4</sup> Domestic fuel prices are therefore set by world markets and the exchange rate, and the subsidy existed to sever that link, holding the pump price below the landed cost of imported fuel.<sup>5</sup> As world prices and the cost of the Naira moved, the fiscal burden of maintaining that gap grew, which is why the subsidy repeatedly returned to the centre of national budget politics.

The subsidy became a fixed point of the social contract. Introduced in the oil-boom years as a way to share resource wealth and stabilize prices, it came to be treated as an

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<sup>4</sup>Private investment is beginning to change this, notably the Dangote refinery, which commenced operations in 2024.

<sup>5</sup>The subsidy kept Nigerian pump prices among the lowest in the world: in early 2024, after the reform, Nigeria still had the 22nd cheapest petrol of 168 countries surveyed (The Punch, 2024). The ranking compares nominal prices and does not adjust for Nigeria's low per-capita income, so it overstates how affordable fuel is for Nigerian households (Africa Check, 2024).

entitlement rather than a discretionary expenditure (Ozili & Obiora, 2023). Successive governments that tried to reduce it met organized resistance. When President Goodluck Jonathan removed the subsidy in January 2012, the ‘Occupy Nigeria’ protests and a national strike shut down economic activity and forced a partial reversal within days (Adebayo, 2015; Egbunike, 2015). The episode set the political backdrop against which the 2023 removal would be judged, and it established that the subsidy could not be withdrawn quietly. That a population this poor would mobilize to defend cheap fuel reflects how little else the state reliably provides (McCulloch, Moerenhout, & Yang, 2021).

Two features of Nigerian life translate a fuel price increase into a broad welfare shock. The first is the electricity crisis. Around 40 percent of households have a grid connection, and those that do experience outages much of the time (Aliyu, Ramli, & Saleh, 2013). The shortfall is met privately: about three-quarters of electricity consumption comes from generators (Nnodim, 2023), and the cost of running them falls hardest on poorer households (Oseni, 2016). For manufacturers, electricity makes up 30 to 35 percent of production costs, far above international norms (Adenikinju, 2005). The second is dependence on road transport. Commercial vehicles, motorcycles, and tricycles run on petrol (Soile, Tsaku, & Musa-Yar’Adua, 2014), and after a price increase the poorest Nigerians spend a large share of their income simply moving themselves and their goods (Ocheni, 2015).<sup>6</sup>

These features define two channels through which removal reaches the household. The first is a production-cost channel. Most productive activity, from a farmer’s irrigation pump to a factory’s assembly line, depends on generator power (Bazilian & Onyeji, 2012). When fuel prices rise, the cost of that power rises with them, and firms operating on thin margins pass the increase forward into the prices of what they sell (Ocheni, 2015). The second is a transportation-cost channel. Higher fuel prices raise the cost of moving goods from where they are produced to where they are sold, which

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<sup>6</sup>The desperation that high fuel prices create was illustrated in October 2024, when a fuel tanker overturned in Jigawa State and exploded as residents rushed to collect the spilled petrol, killing more than 150 people, most of them young (Al Jazeera, 2024).

matters most for food, since production is rural and consumption is increasingly urban (Meludu et al., 2024; Soile et al., 2014).

The two channels compound. A household buying food pays more because the farmer paid more to irrigate, the haulier paid more to carry the harvest, and the trader paid more to keep it cool. Each link adds the fuel price increase to the final retail price. More than half of the welfare impact of a fuel subsidy reduction comes from these indirect effects rather than from households' direct fuel purchases (Arze del Granado et al., 2012), and although removal raises aggregate output, it bears unfavourably on poor households through this transmission (Siddig et al., 2014). The indirect effects are larger in Nigeria because generator dependence embeds the fuel price in almost every good and service, the absence of alternative energy sources limits substitution (Oladosu & Adegbulugbe, 1994), and widespread poverty means a price increase can push a household below a critical consumption threshold (World Bank, 2022). The central expectation of this paper follows: removal of fuel subsidies reduces consumption of basic goods and lowers subjective welfare among Nigerian households.

How the burden falls across areas more and less integrated into formal fuel markets is harder to predict in advance. Greater integration raises direct exposure to the price shock, but the economic development that accompanies fuel infrastructure may also give households more room to adjust, so the net differential is an empirical question.

## **3 Research Design**

### **3.1 Data**

The analysis uses the Nigeria National Longitudinal Phone Survey (NLPS), conducted by the Nigerian Bureau of Statistics with the World Bank, which has tracked about 2,900 households by telephone every three months since January 2021. The NLPS draws its sample from the GHS-Panel and covers all six geopolitical zones; as a telephone survey it under-represents households without mobile-phone access, a limitation common to high-frequency phone surveys in low-income settings. I use Rounds

6, 8, 9, 10, and 12, which span the period before and after the May 2023 removal. Rounds 6 (October 2022) and 8 (April 2023) precede the reform, Round 9 (late May 2023) straddles the inauguration, and Rounds 10 (October 2023) and 12 (July 2024) follow it. Household background characteristics come from the GHS-Panel Wave 4 (2018/2019), which provides covariates fixed well before the reform. Round 12 also fields a module on subsidy awareness and opinion, which I use for the analysis of attitudes. Table 1 reports descriptive statistics.

Table 1: Descriptive statistics

|                                              | Pre-Removal<br>(Round 8) | Post-Removal<br>(Rounds 10, 12) | Difference | Total |
|----------------------------------------------|--------------------------|---------------------------------|------------|-------|
| <i>Panel A: Sample characteristics</i>       |                          |                                 |            |       |
| N                                            | 1,144                    | 2,493                           |            | 3,637 |
| Urban (%)                                    | 42.2                     | 40.3                            |            | 40.9  |
| Female household head (%)                    | 16.2                     | 16.9                            |            | 16.7  |
| Mean household size                          | 6.1                      | 6.2                             |            | 6.2   |
| <i>Panel B: Welfare outcome means</i>        |                          |                                 |            |       |
| Food consumption adequate                    | 0.450                    | 0.278                           | −0.172     |       |
| Clothing consumption adequate                | 0.554                    | 0.474                           | −0.080     |       |
| Housing consumption adequate                 | 0.599                    | 0.579                           | −0.020     |       |
| Healthcare adequate                          | 0.628                    | 0.500                           | −0.128     |       |
| Living standards (1–4 scale)                 | 2.15                     | 1.89                            | −0.26      |       |
| Happiness (1–4 scale)                        | 2.73                     | 2.40                            | −0.34      |       |
| <i>Panel C: Treatment intensity</i>          |                          |                                 |            |       |
| Mean fuel depots per state                   |                          | 3.4                             |            |       |
| States with zero depots                      |                          | 15                              |            |       |
| Max depots (Lagos)                           |                          | 59                              |            |       |
| <i>Panel D: Subsidy awareness (Round 12)</i> |                          |                                 |            |       |
| Aware of fuel subsidy (%)                    | –                        | 50.9                            |            |       |
| Approve of removal (%)                       | –                        | 12.7                            |            |       |

Notes: NLPS analysis sample (N = 3,637). Pre-removal = Round 8; post-removal = Rounds 10, 12. Panel B: binary welfare outcomes coded 1 = adequate, 0 = inadequate; living standards and happiness on 1–4 scales, higher = better. Panel C: fuel depot distribution across states. Panel D: Round 12 subsidy awareness and approval.

The welfare outcomes are six self-assessed measures. Four are binary indicators of whether the household considers its consumption of food, housing, clothing, and healthcare adequate (coded 1 = adequate, 0 = inadequate). Two are ordinal measures of living standards and happiness on four-point scales, recoded so that higher values mean better. These questions appear in Rounds 8, 10, and 12, so the pooled welfare sample has one pre-reform period (Round 8) and two post-reform periods (Rounds 10

and 12), giving 3,637 observations after the merge with baseline covariates. Households that report receiving no healthcare in the previous month are dropped from the healthcare regression, since their standard of care cannot be assessed, which reduces that sample to 2,885. Round 9 straddles the inauguration and is held back for the timing analysis described later. The survey also fields a battery of food security questions, which I analyze in full in the appendix so that the main text can concentrate on subjective welfare and on household attitudes.

Treatment intensity comes from two sources of variation fixed before the reform. The first is the geographic distribution of petroleum depots across states. These depots are the physical infrastructure through which subsidized fuel reached consumers, so states with more depots were more deeply integrated into the formal distribution system and more exposed to the price shock. Depots are concentrated in a few southern states, with Lagos alone holding 59 and fifteen states, mostly in the north, holding none (Appendix Figure A1). Their locations were set decades ago by geology and logistics, not by recent welfare trends. The second is urban–rural classification, which captures differential dependence on the formal fuel market: urban households consume 67 percent of petroleum products despite making up less than half the population (Soile & Mu, 2015).

### **3.2 Empirical Strategy**

A simple before-and-after comparison cannot isolate the effect of subsidy removal, because the Naira devaluation, seasonal variation, and broader macroeconomic developments all moved at the same time. I report such comparisons because they describe what happened to households across the reform, but I treat them as descriptive. To learn how the burden was distributed, I compare how outcomes changed across households that differed in their exposure to the reform, holding constant anything that affected all of them equally (Callaway, Goodman-Bacon, & Sant’Anna, 2024; Roth, Sant’Anna, Bilinski, & Poe, 2023).

The primary specification uses depot intensity as a continuous measure of expo-

sure. For household  $i$  in state  $s$  at time  $t$ ,

$$Y_{ist} = \alpha + \beta_1 \text{Post}_t + \beta_2 \log(\text{Depots}_s + 1) + \beta_3 (\text{Post}_t \times \log(\text{Depots}_s + 1)) + \gamma X_{ist} + \mu_s + \epsilon_{ist}, \quad (1)$$

where  $\text{Post}_t$  marks the post-removal rounds,  $\log(\text{Depots}_s + 1)$  is log fuel depots in the state,  $X_{ist}$  are household controls (household size, head gender, urban status),  $\mu_s$  are state fixed effects, and standard errors are clustered at the local government area (LGA) level. Because depot counts vary only across states, the level term  $\beta_2$  is absorbed by the state fixed effects. The coefficient of interest is  $\beta_3$ , which measures how the change from before to after the reform differed between high- and low-depot states.

I interpret  $\beta_3$  with caution. Depot count is not a clean measure of fuel-market exposure alone. States with more depots are also more urbanized, more industrialized, and more economically diversified, so the interaction captures the combined influence of fuel-market integration and economic development. The two pull in opposite directions: integration should deepen exposure to the price shock, while development should provide more room to absorb it. The estimate recovers their net effect, not the effect of fuel exposure holding development constant. I therefore present these results as evidence on how the burden was distributed across more- and less-developed states, not as the causal effect of fuel-market exposure. The identifying assumption for the differential is that, absent the reform, welfare in high- and low-depot states would have moved in parallel, which I assess with the event study below.

Two further designs probe the same question and are reported in the appendix. The urban–rural design replaces depot intensity with an urban indicator; its predicted sign is ambiguous, because rural households also rely heavily on generators and fuel-intensive transport (Berkouwer & Dean, 2022). The triple-difference design asks whether urban households in high-depot states changed by more than the separate urban and depot patterns would predict (Olden & Møen, 2022). It requires only that the urban–rural gap would have evolved similarly across high- and low-depot states absent the

reform, an assumption weaker than parallel trends in levels and one that tolerates shocks like the devaluation as long as they did not interact with depot intensity.

I assess parallel trends and robustness in several ways. Event study plots trace mean welfare for high- and low-depot states and for urban and rural households across rounds, so that pre-reform similarity can be inspected directly. Because the welfare questions appear in only one pre-reform round (Round 8), a formal pre-trend test is not feasible, and the event study evidence is necessarily visual (Rambachan & Roth, 2023; Roth, 2022). I also exploit the timing of Round 9, fielded around the 29 May inauguration: interview dates were set by survey logistics rather than household characteristics, so comparing households interviewed just before and just after the announcement gives a regression-discontinuity-in-time estimate of any immediate effect (Hausman & Rapson, 2018), though the small number of post-inauguration interviews limits its power. Finally, I re-estimate the binary outcomes with logit models, cluster standard errors at the state level, and run a placebo test on household size, which fuel prices should not move.

## **4 Results**

### **4.1 Welfare Fell After the Reform**

Welfare declined across most dimensions between the pre- and post-reform rounds. Table 2 reports the average change in each outcome. Food consumption adequacy fell by about 17 percentage points and clothing adequacy by about 8. Self-assessed health-care adequacy fell by about 13 percentage points. Living standards and happiness, on four-point scales, both declined. Housing adequacy was the exception, moving in the same direction by an amount indistinguishable from zero. Five of the six measures worsened, and the estimates are statistically significant at conventional levels. These comparisons cannot separate the reform from the concurrent devaluation and inflation, so I interpret them as a description of household conditions across the reform rather than as its isolated causal effect. The appendix traces food prices, energy prices,

and the Naira-dollar exchange rate over the period, each of which rose around the inauguration. The direction and breadth of the decline are consistent with the expectation that higher fuel prices transmit through production and transportation costs to household welfare.

Table 2: Pre/post change in welfare outcomes

|                | (1)<br>Food            | (2)<br>Housing      | (3)<br>Clothing        | (4)<br>Health          | (5)<br>Living          | (6)<br>Happiness       |
|----------------|------------------------|---------------------|------------------------|------------------------|------------------------|------------------------|
| Post           | −0.1739***<br>(0.0158) | −0.0176<br>(0.0171) | −0.0779***<br>(0.0172) | −0.1257***<br>(0.0202) | −0.2519***<br>(0.0304) | −0.3415***<br>(0.0304) |
| Controls       | Yes                    | Yes                 | Yes                    | Yes                    | Yes                    | Yes                    |
| State FE       | Yes                    | Yes                 | Yes                    | Yes                    | Yes                    | Yes                    |
| Clustered SE   | LGA                    | LGA                 | LGA                    | LGA                    | LGA                    | LGA                    |
| Observations   | 3,637                  | 3,637               | 3,637                  | 2,885                  | 3,637                  | 3,637                  |
| R <sup>2</sup> | 0.068                  | 0.024               | 0.032                  | 0.035                  | 0.056                  | 0.098                  |

Notes: OLS with state FE. SEs clustered at LGA level. (1)–(4): 1 = adequate, 0 = inadequate; (5)–(6): higher = better. Controls: household size, head gender, urban. Healthcare (4) drops households not receiving care in the past month, reducing N. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 4.2 How the Burden Was Distributed

The decline was uneven across states. Before turning to the estimates, Figure 1 plots all six welfare outcomes across rounds for states with above- and below-median depot intensity. In the single available pre-reform period (Round 8) the high- and low-depot lines are close for every outcome, and they separate only after May 2023, with the high-depot line ending above the low-depot line for food and clothing. With one pre-reform period a formal pre-trend test cannot be run (Roth, 2022), but the visual evidence is consistent with parallel trends. The same plot for urban and rural households (Appendix Figure A2) shows the two groups moving together throughout, without a post-reform divergence.

Table 3 reports the depot intensity DiD results. The Post coefficient repeats the broad decline. The interaction of Post with log depots is positive and statistically significant for food consumption, clothing, and happiness, meaning that states with more depot infrastructure recorded *smaller* declines in these outcomes. Housing, healthcare,

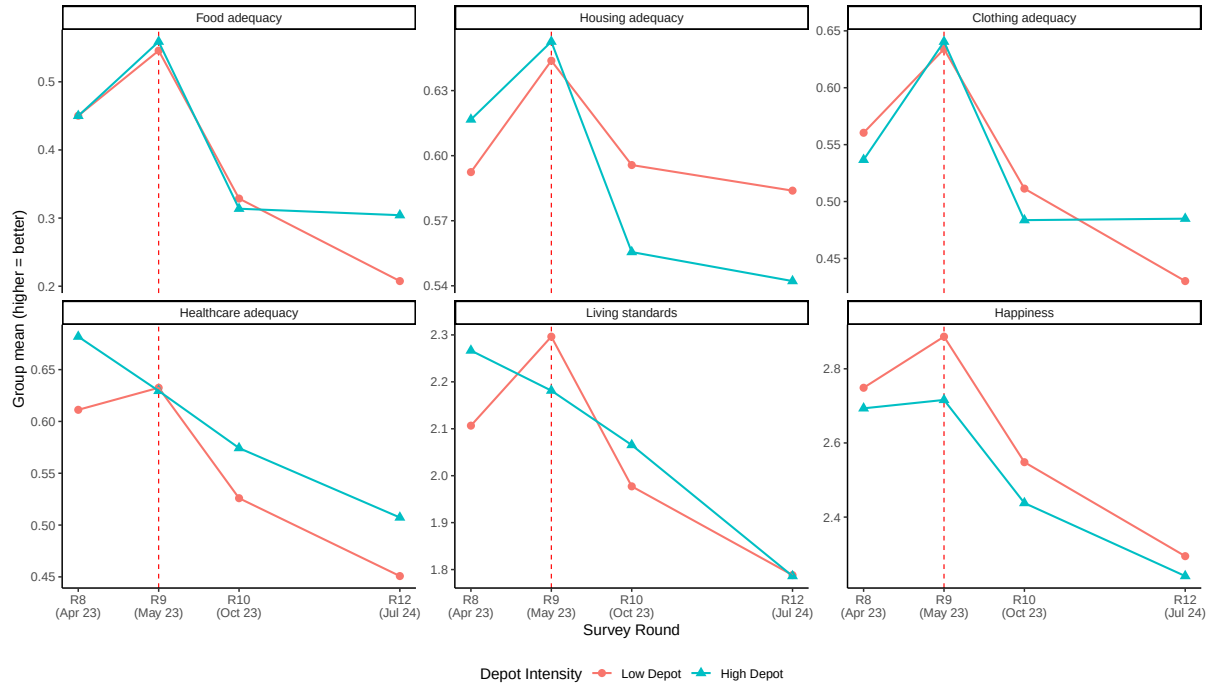


Figure 1: Event study: all six welfare outcomes by fuel depot intensity across survey rounds. Binary outcomes (food, housing, clothing, healthcare) are the proportion reporting adequacy; living standards and happiness are 1–4 scales. Higher is better throughout. High depot = states with above-median depot counts. The dashed line marks fuel subsidy removal (29 May 2023).

and living standards show no differential. The same is observed at the state level (Appendix Figure A3): the change from before to after the reform rises with depot count for food, clothing, and happiness, and is weak or flat for the other three outcomes.

Depot infrastructure trends with economic development, denser supply chains, and more margins for adjustment, and on these broad welfare measures that resilience appears to outweigh the deeper fuel-market exposure that the same infrastructure brings. The result suggests that better-developed states absorbed the shock more easily; it does not isolate a protective effect of fuel-market integration, since the two move together. I return to this limitation in the discussion.

The urban–rural design (reported in Appendix Table A2) shows an interaction indistinguishable from zero for every outcome except housing, where it is marginally positive. Urban and rural households followed similar welfare paths, which fits a setting where rural households depend on generators and fuel-intensive transport almost as much as urban ones, and indicates that the urban–rural split is too coarse to

Table 3: Fuel depot intensity DiD: welfare outcomes

|                      | (1)<br>Food            | (2)<br>Housing      | (3)<br>Clothing        | (4)<br>Health          | (5)<br>Living          | (6)<br>Happiness       |
|----------------------|------------------------|---------------------|------------------------|------------------------|------------------------|------------------------|
| Post                 | −0.2104***<br>(0.0207) | −0.0141<br>(0.0228) | −0.1088***<br>(0.0226) | −0.1383***<br>(0.0273) | −0.2494***<br>(0.0405) | −0.4062***<br>(0.0400) |
| Post × log(Depots+1) | 0.0408***<br>(0.0140)  | −0.0040<br>(0.0168) | 0.0346**<br>(0.0157)   | 0.0156<br>(0.0261)     | −0.0027<br>(0.0239)    | 0.0724**<br>(0.0282)   |
| Controls             | Yes                    | Yes                 | Yes                    | Yes                    | Yes                    | Yes                    |
| State FE             | Yes                    | Yes                 | Yes                    | Yes                    | Yes                    | Yes                    |
| Clustered SE         | LGA                    | LGA                 | LGA                    | LGA                    | LGA                    | LGA                    |
| Observations         | 3,637                  | 3,637               | 3,637                  | 2,885                  | 3,637                  | 3,637                  |
| R <sup>2</sup>       | 0.069                  | 0.024               | 0.033                  | 0.035                  | 0.056                  | 0.100                  |

Notes: DiD with depot intensity as continuous treatment. (1)–(4): 1 = adequate, 0 = inadequate; (5)–(6): higher = better. SEs clustered at LGA level. Key coefficient: Post × log(Depots+1). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

capture fuel-market exposure. The triple-difference design (Appendix Table A3) is statistically significant only for living standards, where urban households in high-depot states held up better than the separate patterns predict, echoing the depot result; the other five outcomes show nothing. Both designs point the same way as the depot estimates: more-developed states, proxied by fuel infrastructure, absorbed the shock with smaller declines.

### 4.3 How Households Judged the Reform

The survey lets households speak for themselves, and what they say is reported and discussed here. About half of households were aware of the fuel subsidy, and of those who were aware, nearly all knew it had been removed (Figure 2). Approval for the reform was rare. Of households that gave a view, 87 percent disapproved of the removal (Table 1, Panel D). The disapproval was not confined to any one region or to urban households; it was close to universal among those with an opinion.<sup>7</sup>

The reasons households gave are reported in Figure 3. Among the minority who approved, the most common reasons were that the subsidy mainly benefited the wealthy, that the funds could be better used elsewhere, and that the subsidy regime was corrupt. These are, almost exactly, the three arguments economists make against fuel sub-

<sup>7</sup>This opposition predates the 2023 reform: survey evidence shows Nigerians opposed subsidy removal well before it was enacted (McCulloch et al., 2021).

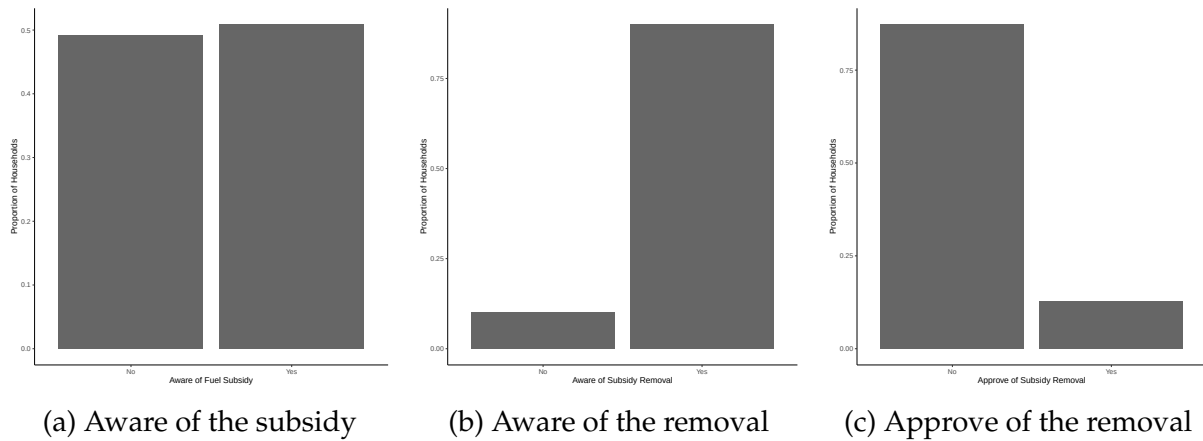


Figure 2: Household awareness of the fuel subsidy and approval of its removal (Round 12). Awareness of the removal (Panel b) is asked only of households aware of the subsidy. Bars show the percentage of households giving each response.

sidies: regressivity, misallocation, and rent-seeking (see Section 2.2). Among the majority who disapproved, the reasons were different in kind. More than three-quarters objected because removal had raised the prices of goods, and most of the remainder because it had harmed the poor. The public rejected the reform on the basis of its lived consequences, not its principle.

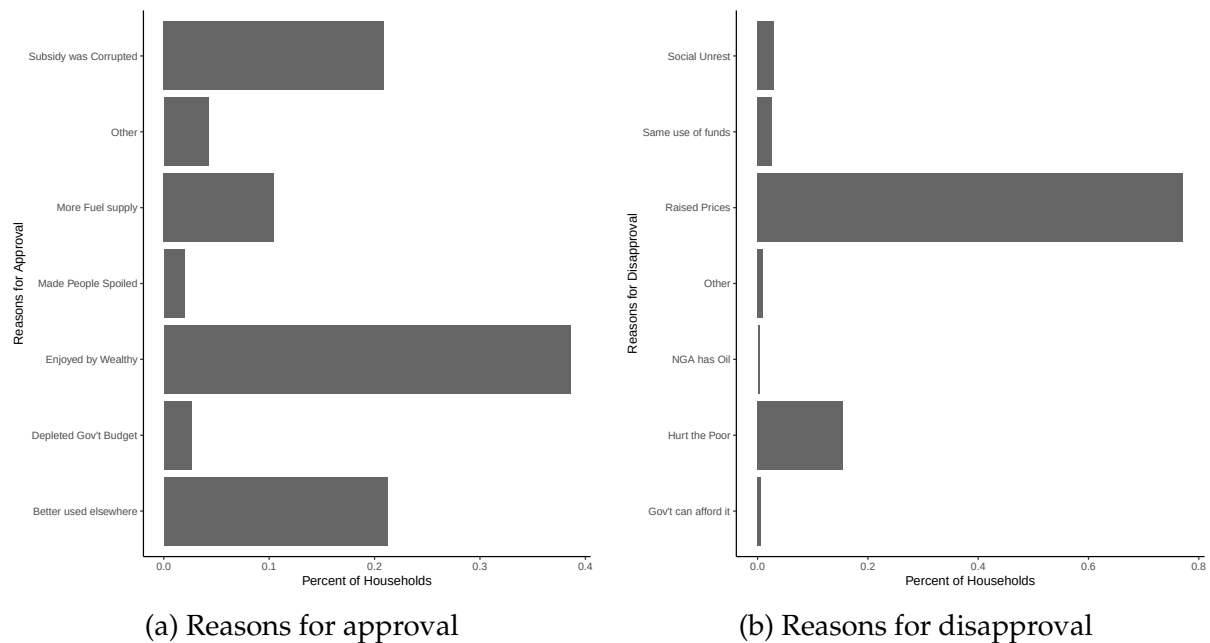


Figure 3: Household reasons for approving or disapproving of fuel subsidy removal (Round 12). Bars show the share of respondents giving each reason, among those who approved (Panel A) and those who disapproved (Panel B).

This gap matters for the politics of reform. The efficiency and fairness arguments

that justify removal are understood by part of the public, but they are dominated by the immediate experience of higher prices. A reform that delivers its costs at once and its benefits later, through reallocated fiscal resources, faces a constituency that judges it by the costs. The near-universal disapproval recorded here is the same force that reversed the 2012 attempt, and it conditions whether any future reform can hold.

#### **4.4 Robustness Checks**

The welfare losses did not appear on the day of the announcement. Comparing households interviewed just before and just after the 29 May inauguration within Round 9 (Appendix Table A9) gives coefficients that are small and indistinguishable from zero for every outcome except happiness, which shows a marginal decline. This fits an account in which losses operated through indirect channels, production and transportation costs and supply-chain adjustment, that take time to work through rather than registering immediately. The small sample ( $N = 1,249$ ) also limits what this test can detect. The timing carries a policy implication that I take up in Section 5.2: the loss accumulated over months, so an interval existed in which compensation could have reached households ahead of the price pass-through.

The heterogeneity results survive alternative specifications. Logit estimates of the binary outcomes (Appendix Tables A4 and A5) reproduce the direction of the OLS results: the depot interaction is positive and statistically significant for food consumption and clothing, and the urban interaction is positive and marginal for housing. The findings also hold when standard errors are clustered at the state level. A placebo test shows no differential pre-post change in household size across treatment groups, which supports the validity of the identifying variation.

## 5 Discussion and Conclusion

### 5.1 Discussion

Household welfare fell after the May 2023 reform on five of six self-assessed dimensions, with no compensating support, and the decline was still present in July 2024, fourteen months later. The decline was smaller in the more economically developed states that fuel infrastructure marks out. Households that lived through the reform rejected it almost unanimously, on the basis of the prices they faced.

The welfare evidence rests on a nationally representative panel observed before and after the reform, and it holds across six measures and several specifications. It is not the isolated causal effect of removal: the Naira devaluation and broader inflation moved at the same time, and no untreated comparison group exists. What the data establish is that households were worse off across the reform, by their own assessment, on food, clothing, healthcare, living standards, and happiness alike.

The heterogeneity results suggest that economic development, which fuel infrastructure proxies, buffered the shock. States with more depots recorded smaller declines in food adequacy, clothing, and happiness, and urban households in those states held their living standards better than the lower-order patterns predict. Because depot count and development cannot be separated in these data, I cannot attribute the buffering to fuel-market integration specifically, and the average welfare cost, which every household bore, is larger than these differences across states. On the food security measures in the appendix the gradient reverses: the more integrated states fared worse on specific dietary behaviours, so the distribution of the burden depends on which aspect of welfare is measured.

A minority of Nigerians articulated the economic case for removal, citing the subsidy's capture by the wealthy, its fiscal cost, and its corruption (Arze del Granado et al., 2012; Harberger, 1964). The majority judged the reform by its effect on prices. A reform whose costs arrive at once and whose benefits depend on the later use of freed resources needs visible compensation to hold, and the 2023 removal offered none.

Several limitations bound these conclusions. Each outcome has one pre-reform period, so parallel trends rest on visual event study evidence. Treatment is measured at the state level, and the depot measure is confounded with development. The before-and-after and difference-in-differences estimates are descriptive and differential respectively; neither recovers the average causal effect. The opinion module was fielded once, so I observe the distribution of opinion at one point and not its movement.

## 5.2 Policy Implications

Nigeria removed the subsidy with no compensation in place, no published account of what the savings would fund, and no rule governing the fuel price that followed. The reform literature identifies each as a condition for success (Clements et al., 2013). Five recommendations follow from the results above read against what Nigeria has done since May 2023.

*Deliver compensation before prices rise, and sustain it.* The Renewed Hope conditional cash transfer, which pays 75,000 naira in three tranches, was relaunched in October 2023, five months after the price increase. Two years after the reform, 5.6 million households, about 37 percent of the 15 million target, had received at least one tranche, and fewer than 1.3 million a third (World Bank, 2025). The welfare declines measured here run to July 2024, so most target households received nothing over the period in which the loss was recorded. Welfare did not fall on the day of the announcement (Section 4.4); it accumulated as fuel costs passed through production and transport into retail prices over the following months. A transfer despatched in the weeks after the announcement would have reached households ahead of most of that loss, which requires a household register and a payment system in operation before the reform is announced. The declines had not closed after fourteen months, so a three-tranche payment is calibrated to a shorter disturbance than these data show.

*Weight transfers toward the least developed states.* The broad welfare declines were largest in the states with the least fuel infrastructure, which are the least urbanized and least diversified and lie mostly in the north. Targeting depends on where the

losses were largest; the confound between fuel-market integration and development concerns why, and does not change the targeting conclusion. A transfer of uniform value per household under-compensates these states. The food security results point the other way: on dietary behaviours, the more integrated states recorded the larger declines (Appendix Tables A6 to A8). General income support weighted toward less developed states and food-specific support directed at the more integrated ones address different parts of the shock. State-level weighting is second-best to a complete household register, which Nigeria did not have at the time of the reform (Porisky, Mohamed, & Muthui, 2023), and can be withdrawn as the register improves.

*Publish the fiscal savings and their uses.* About half of households knew a fuel subsidy existed. Of those with a view, 87 percent disapproved of removal, most citing prices; the minority who approved cited regressivity, alternative uses of the money, and corruption, the three arguments the economics literature makes. Part of the public therefore held the case for reform, and the government did not communicate it. Removal saved an estimated 2.6 percent of GDP in 2024, but the national oil company began transferring the gains to the Federation Account only in January 2025 and has remitted about half, applying the rest to arrears (World Bank, 2025). The expected savings and their uses should be published before removal, the realized savings routed through the Federation Account in full, and the resulting spending reported against the original commitment.

*Lower the fuel intensity of household consumption.* About three-quarters of electricity consumption comes from private generators, and more than half the welfare impact of a fuel subsidy reduction operates through indirect price channels (Arze del Granado et al., 2012). This is why the declines reached food, clothing, and healthcare. Grid reliability and fuel switching in transport, such as the Presidential Compressed Natural Gas Initiative established in 2023 (Presidential Initiative on Compressed Natural Gas, 2024), reduce that pass-through for the next price movement. They are slow instruments and do not substitute for compensation.

*Replace the administered price with a rule.* Nigeria imports the fuel it consumes, so

the domestic cost tracks world prices and the exchange rate. Holding the pump price below that cost made the fiscal burden a function of variables the government does not control, which is why the subsidy returned to budget politics repeatedly and why the 2012 attempt was reversed within days. An automatic pricing mechanism with a published formula and a smoothing band converts a recurring political decision into a rule (Clements et al., 2013), and a transfer that scales with the price gives households a stated expectation of what follows when world prices rise.

This paper does not estimate the effect of any of these instruments. The data support the diagnosis that the losses were broad, persisted beyond a year, fell hardest on the least developed states, differed across outcomes, and were rejected by a public that a compensation and disclosure package addresses directly. The policies suggested here reflect what the literature has found to be useful in other places.

### **5.3 Conclusion**

Future research can use later NLPS rounds to show whether the declines fade or persist as households adjust, which determines how long compensation should run. The incomplete and staggered rollout of the Renewed Hope transfer supplies the comparison group this paper lacks: households reached early, reached late, and not yet reached differ in the timing of a payment whose allocation was governed by register coverage and verification and is unrelated to welfare trends, and a design exploiting that variation could identify the effect of compensation on welfare and on approval. Household-level measures of fuel intensity from the GHS consumption modules, or market-level price data matched to depot geography, would separate fuel-market integration from development and estimate pass-through directly. Consumption expenditure data would give the change in poverty headcount and depth against which the cost of alternative compensation packages could be weighed. Repeated opinion modules would show whether approval recovers as savings become visible, and an information provision experiment would test whether the gap between the case for reform and the public verdict is a gap in information or in interests.

Nigeria removed a subsidy that was inefficient, regressive, and captured, and did so with no compensation in place, no account of where the money would go, and no rule to govern the price that followed. Household welfare fell across five of six dimensions and had not recovered fourteen months later, the losses were heaviest in the states least equipped to absorb them, and the public rejected the reform for reasons rooted in the prices they paid. Whether a future removal holds will depend on what is delivered alongside it.

## **Data Availability Statement**

The Nigeria National Longitudinal Phone Survey data analysed in this study are publicly available from the World Bank Microdata Library, as are the GHS-Panel Wave 4 data used for baseline covariates. The fuel depot counts are derived from the published NUPRC depot registry. A replication package containing the data preparation and analysis code will be deposited in the Harvard Dataverse. To preserve anonymity during review, the repository citation is omitted here and will be supplied on acceptance.

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## **Conflict of Interest Statement**

The author declares no conflict of interest.

## **Ethics Statement**

This study is a secondary analysis of de-identified, publicly available survey data and did not require ethical approval.

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